Unit: Unit 1: Ratio, Rate, Percentage
Topic: Ratios Proportions

Name: ________________
Date: __________

Foundation (Jalapeno)

Q1. A space station has 3 : 5 oxygen to nitrogen ratio. If there are 15 units of oxygen, how many units of nitrogen are there?
(A) 5
(B) 9
(C) 10
(D) 12
(E) 15

Q2. A planet's mass to volume ratio is 4 : 7. If the mass is 16, what is the volume?
(A) 20
(B) 21
(C) 28
(D) 32
(E) 35

Q3. In a lab experiment, 2 grams of substance A reacts with 5 grams of substance B. How many grams of B will react with 8 grams of A?
(A) 10
(B) 15
(C) 20
(D) 25
(E) 30

Q4. A spaceship travels 240 miles in 4 hours. If it travels for 6 hours, how many miles will it cover?
(A) 280
(B) 300
(C) 320
(D) 360
(E) 380

Q5. The ratio of the radius of the Earth to the radius of the Moon is 11 : 3. If the radius of the Earth is 3960 miles, what is the radius of the Moon?
(A) 900
(B) 1080
(C) 1200
(D) 1300
(E) 1400

Q6. A scientist mixes 2 parts of a chemical with 3 parts of water. If she uses 12 parts of the chemical, how many parts of water should she use?
(A) 12
(B) 15
(C) 16
(D) 18
(E) 20

Q7. A space probe's fuel consumption is in the ratio 5 : 8 of distance traveled. If it travels 40 miles, how much fuel will it consume?
(A) 24
(B) 30
(C) 32
(D) 40
(E) 50

Q8. The ratio of the number of scientists to engineers in a space agency is 7 : 9. If there are 63 scientists, how many engineers are there?
(A) 70
(B) 75
(C) 81
(D) 90
(E) 100

Open Ended Questions (FRQ)

Q9. A space station has two air tanks. Tank A has a ratio of oxygen to nitrogen of 2 : 3 and Tank B has a ratio of 3 : 5. If Tank A has 12 units of oxygen, how many units of nitrogen are there in Tank B if it has 15 units of oxygen? Show your work and explain your answer.

Q10. A planet's atmosphere is made up of gases in the ratio 4 : 7 : 10 of nitrogen, oxygen, and carbon dioxide. If a sample of the atmosphere contains 16 units of nitrogen, how many units of oxygen and carbon dioxide are there? Use cross multiplication to solve the problem and show your work.

Chef's Tips

- Tip 1: Ensure students show their work and explain their answers for FRQs
- Tip 2: Encourage students to use cross multiplication to solve proportion problems
- Tip 3: Remind students to check their units and ensure they are consistent throughout the problem